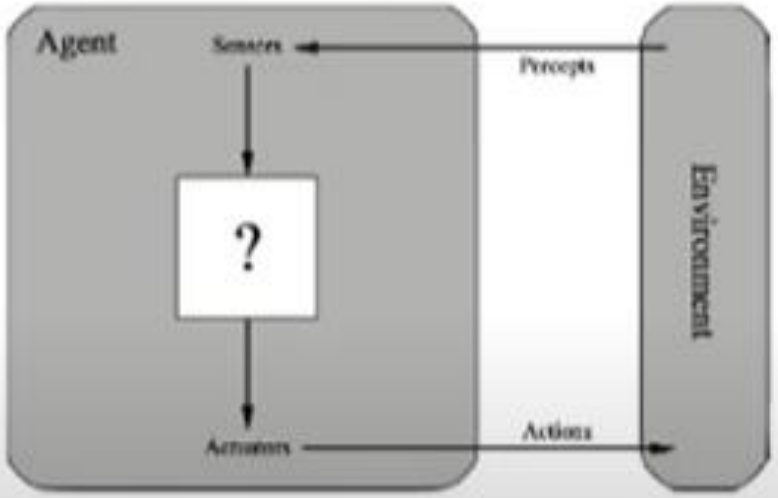
Intro to Multi Agent Systems

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Agents and environments

- **Agent:** An agent is anything that can be viewed as:
  - perceiving its environment through sensors and
  - acting upon that environment through actuators.



- An agent program runs in cycles of: (1)perceive, (2)think, and (3)act.
- **Agent = Architecture + Program**

Agents and environments

- **Human agent:**
  - Sensors: eyes, ears, and other organs.
  - Actuators: hands, legs, mouth, and other body parts.
- **Robotic agent:**
  - Sensors: Cameras and infrared range finders.
  - Actuators: Various motors.
- **Agents everywhere!**
  - Thermostat
  - Cell phone
  - Vacuum cleaner
  - Robot
  - Alexa Echo
  - Self-driving car
  - Human
  - etc.

Well-behaved agents

Rational Agent:

"For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure*, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has."*

Rationality

- Rationality is relative to a performance measure.
- Judge rationality based on:
  - The performance measure that defines the criterion of success.
  - The agent prior knowledge of the environment.
  - The possible actions that the agent can perform.
  - The agent's percept sequence to date.

PEAS

- When we define a rational agent, we group these properties under **PEAS**, the problem specification for the task environment.
- The rational agent we want to design for this task environment is the solution.
- PEAS stands for:
  - **P**erformance
  - **E**nvironment
  - **A**ctuators
  - **S**ensors

PEAS

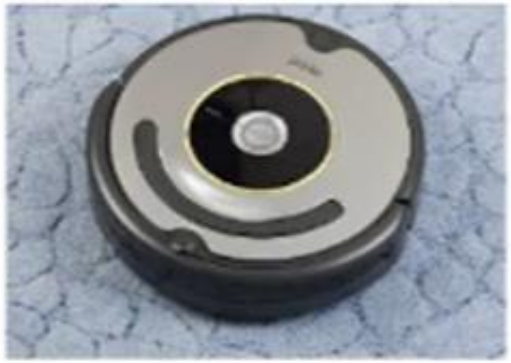
What is PEAS for a self-driving car?



- **P**erformance: Safety, time, legal drive, comfort.
- **E**nvironment: Roads, other cars, pedestrians, road signs.
- **A**ctuators: Steering, accelerator, brake, signal, horn.
- **S**ensors: Camera, sonar, GPS, Speedometer, odometer, accelerometer, engine sensors, keyboard.

PEAS

How about a vacuum cleaner?



iRobot Roomba series

- **P**erformance: cleanness, efficiency: distance traveled to clean, battery life, security.
- **E**nvironment: room, table, wood floor, carpet, different obstacles.
- **A**ctuators: wheels, different brushes, vacuum extractor.
- **S**ensors: camera, dirt detection sensor, cliff sensor, bump sensors, infrared wall sensors.

Environment types

- **Fully observable (vs. partially observable):** An agent's sensors give it access to the complete state of the environment at each point in time.

- **Deterministic (vs. stochastic):** The next state of the environment is completely determined by the current state and the action executed by the agent. (If the environment is deterministic except for the actions of other agents, then the environment is strategic)

- **Episodic (vs. sequential):** The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.

Environment types

- **Static (vs. dynamic):** The environment is unchanged while an agent is deliberating. (The environment is semi-dynamic if the environment itself does not change with the passage of time but the agent's performance score does.)
- **Discrete (vs. continuous):** A limited number of distinct, clearly defined percepts and actions. E.g., checkers is an example of a discrete environment, while self-driving car evolves in a continuous one.
- **Single agent (vs. multi-agent):** An agent operating by itself in an environment.
- **Known (vs. Unknown):** The designer of the agent may or may not have knowledge about the environment makeup. If the environment is unknown the agent will need to know how it works in order to decide.

A MAS has the following characteristics:

- Decentralisation: There is no designated controlling agent. The agents coordinate through observing others, communications, and system-level policies.
- Autonomy: The agents are at least partially independent, self-aware, autonomous. The tasks/goals of the agents are not necessarily the same.
- Local views: No agent has a full global view of the system, or the system is too complex for an agent to make practical use of such knowledge.



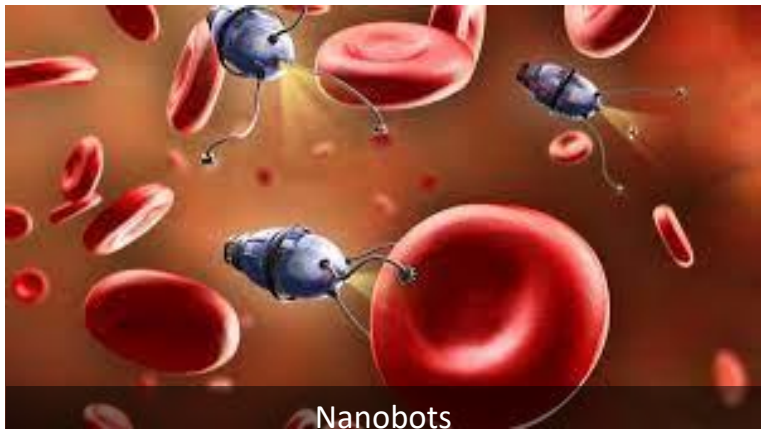
Examples of Multi Agent Systems Agent Systems (MAS)

Robotics

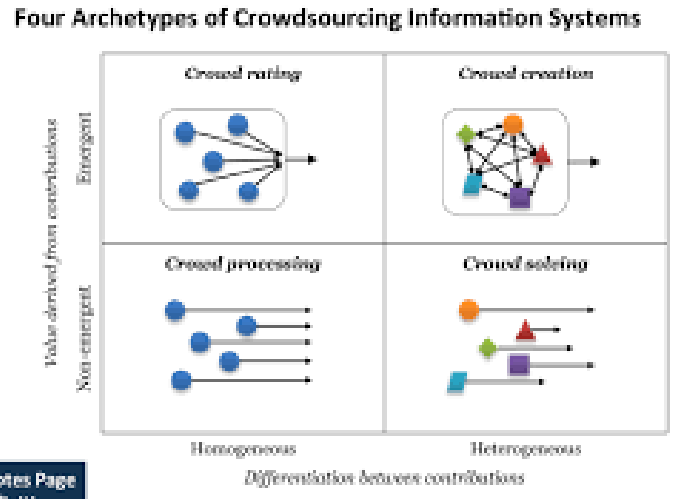




Autonomous Driving



Nanobots



Crowd Sourcing

Crowd Simulations



Investigations of MAS are driven by the following questions:

- **Distributed problem solving**: Designing a MAS for complex computational problems.

  Decompose the problem so that mutli-agents are utilised to find robust and effective solutions to a problem.

- **Coordination and cooperation**:

  How do agents with heterogeneous goals work together? How do they reach consensus? How do they distribute resources?

- **Multi-agent learning**:

  How do agents improve themselves through interacting with other agents? How do social patterns emerge?

Multi-agent System

Main challenge:

- Autonomous control.
- Self-centred agents.
- Partial information.
- A protocol specifies how an agent should act and communicate with other agents. Our challenge is to design protocols that enable the agents to achieve coordination.

In this lecture we will discuss two scenarios that face such issues.

Mutli-agent Pathfinding

Collision avoidance of mobile robots. Imagine mobile robots which are able to move from place to place and pick up or drop off objects. When several robots are placed in the same environment, they may interfere with each other. We need to design mechanisms that ensure the robots do not collide.





- **Centralised control** (Distributed system)
- **Autonomous control** (Multi-agent system)

Disadvantage: High computational cost. We desire a solution that combines both methods: maintain the autonomy of agents, while allowing efficient coordination as in a centralised situation.





Multi Agent Path Finding Challenges ?

Questions. The problem raises several issues:

- How can the agents avoid collision?
- How can they reach goal positions?
- How can they do this efficiently?

What capabilities do the agents have?

- Can they sense their environment?
- Can they control their speed?

What does the environment (V, E) look like?

Hamiltonian circuit

- → **Every node should be visited once and must be once, in a constant speed. Once the route is finished, again, reach to the same starting point and re-iteration of the same path occurs.**

Traffic Law 1

Assumptions:

- Agents have no sensory capability.
- Agents have uniform speed (1 unit of time edge)





Traffic Law 1

Each robot is required to move constantly. The d fixed as follows:

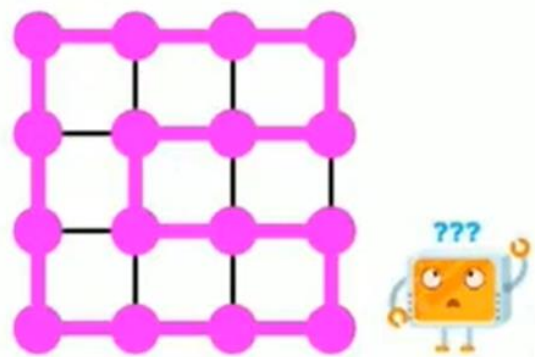
1. Identify a Hamiltonian circuit of the graph
2. From every node, an agent is only able to m in this Hamiltonian path.

Theorem

If G contains a Hamiltonian path, Traffic Law 1 guarantees that no collision will occur, and that each agent will reach each goal in $O(|V|)$ steps.

Problems with Traffic Law 1

- TL1 completely dictates agent behaviours, with no room for individual planning.
- "Rigid laws lead to suboptimal behaviours" – slow arrival time, high energy consumption, especially when $|V|$ is large.



Traffic Law 2

Assumptions:

- Agents have no sensory capability.
- Agents have ability to change speed. So traversing an edge (u, v) may be a continuous movement from u to v within t time units, for arbitrary $t \in \mathbb{R}$.

Question. What is the best way to avoid traffic accidents?



Traffic Law 2

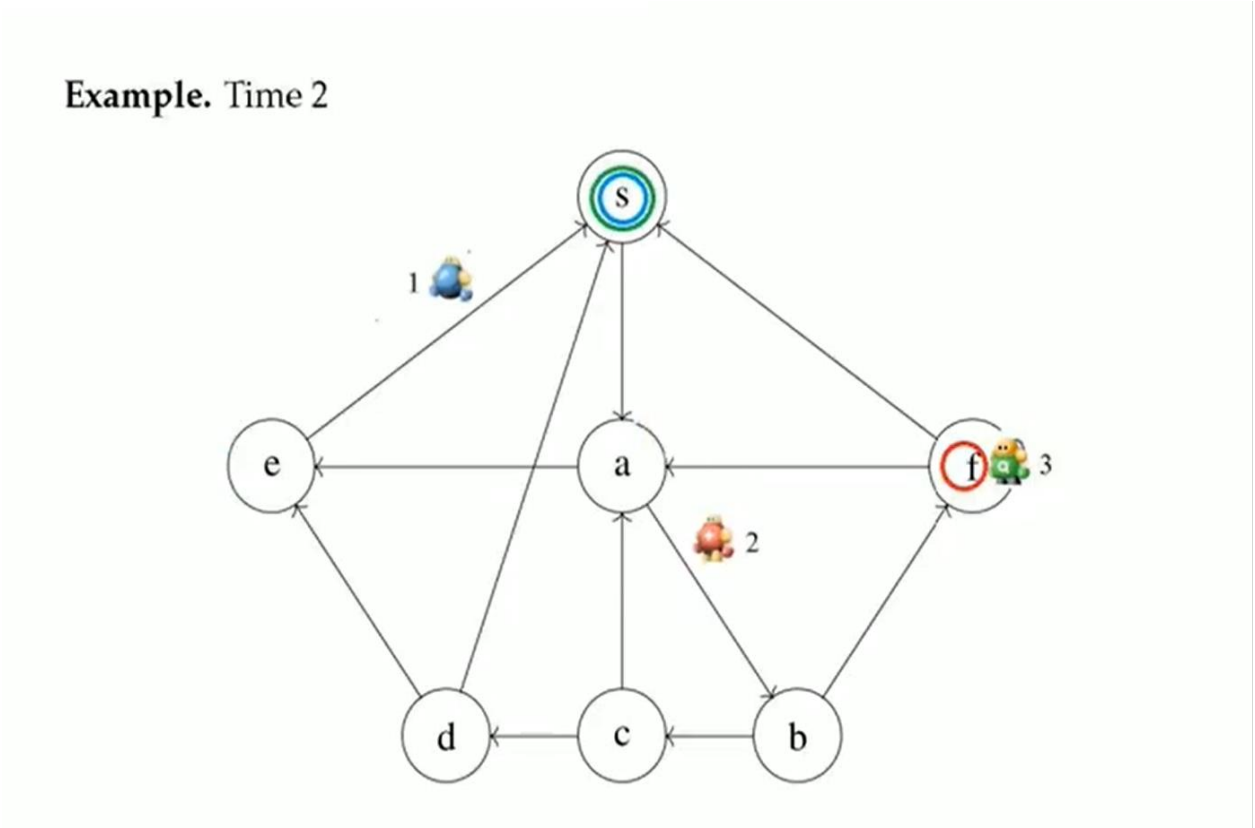
- Order all the agents $a_1, a_2, \ldots, a_m$.
- The agent a_i starts moving at time unit i.
- Each agent moves so slow that traversing each link takes exactly m time units.

Traffic Law-02

Let`s assume, there are 03 different agents as, A, B and C. A starts at time T=A and moves in V_A speed. Similarly, B starts at time T=B and moves in V_B speed and C starts at time T=C and moves in V_c speed. Different time, speed and displacement combinations can be arranged in such a state to avoid collisions.

As in Traffic law 1, here the path is not fixed. Shortest path also can be considered as a parameter along with the other established parameters combinations to determine the collision free overall plan.

The plan should prevent, two agents arriving at the same node at the same time.



More options are provided in the physical ways, allowing choices for the agents

Traffic Law 3

Assumptions:

- Agents have sensory capability of close-by nodes.
- Agents have uniform speed (1 unit of time to travel through an edge)

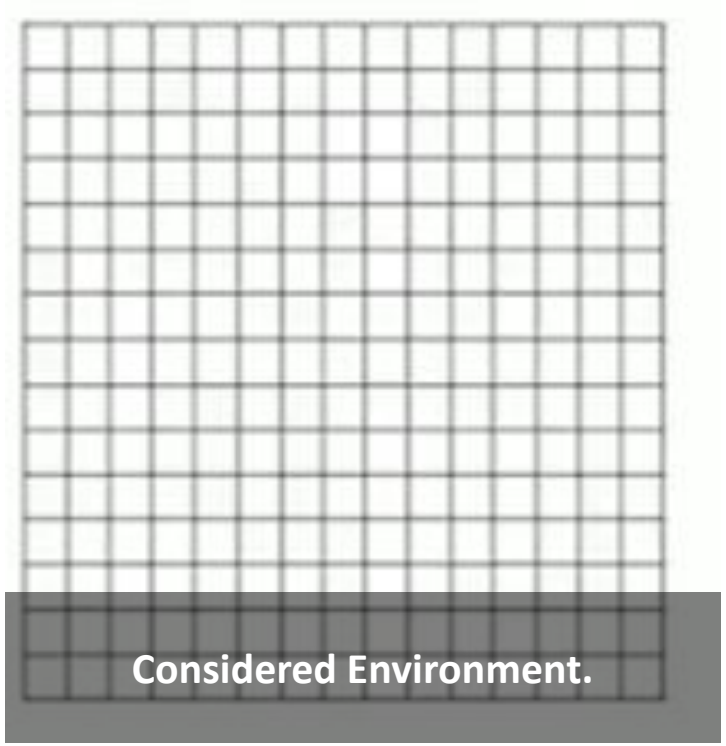




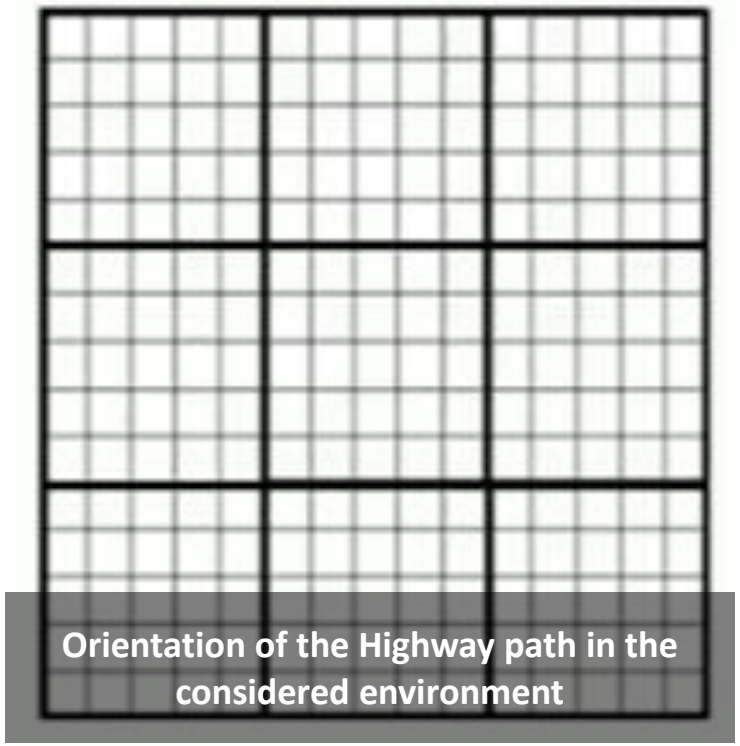


- To speed up travel: Implement “high ways”
- To avoid collisions: Implement “give-way rules”
- To ensure arrival: Implement “local suburbs”

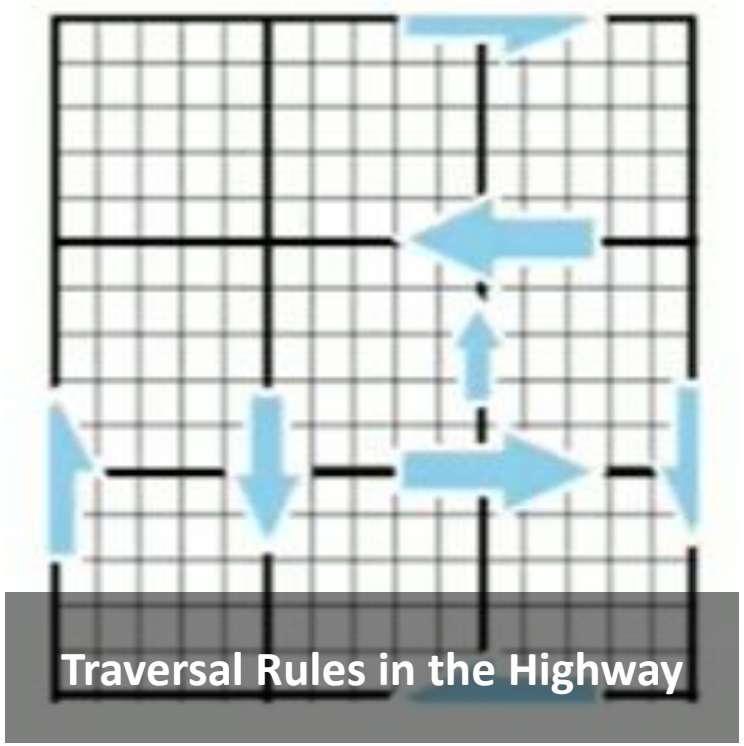
Implementing Highways.



Considered Environment.



Orientation of the Highway path in the considered environment



Traversal Rules in the Highway

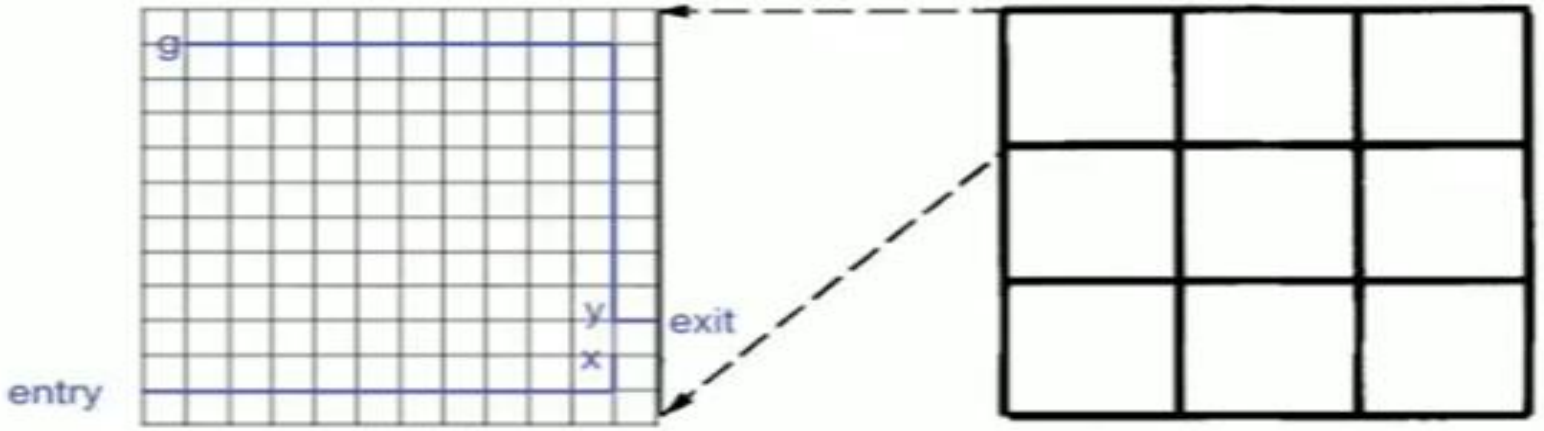
Junction Handling.



Traffic Law 3 (Part II Give-way)

- When an agent enters a junction on the highway, it must wait for agents who also want to enter the junction but arrived there earlier.
- If two agents arrive there together, agents on rows must give way to agents on columns.

Suburban Pathways



Traffic Law 3 (Part III Suburban)

- The agent only enters an internal area from the entry and moves along the blue line to x.
- If y is unoccupied or some one is moving on y, the agent then moves from x to g by moving only up and left (not touching the blue line); otherwise it waits at x for 1 time unit.
- The agent then moves along the blue line to exit.
- The exit is considered a junction on the highway.
- The agent cannot stop unless agents in front does not move.

Reflection on traffic laws

From traffic laws 1 to 3, autonomy has been increased, by providing more options to the agents.

The issue associated with too many options is, operational jeopardy.

When too many options are introduced, governing protocols are also needed.

That`s why, in the traffic law 3, along with more options, more governing protocols are also introduced.

Agents have the autonomy to select the option. But, when an option is selected, they should adhere to the governing protocol applied.